

Hematology Transfusion / BMT / Cellular Therapy Codex Guide

- Block: Hematology
- Method: Codex-authored manual distillation from extracted lecture markdown in this workspace.
- Sources:
 - 20220307_Yui_Transfusion_Medicine.pdf
 - 20230315_Jones_Bone_Marrow_Transplant.pdf
 - 2026_imus_CAR-T_BsAbs.pdf
 - OSFM Heme Workshop Small Group 2025.pdf

One-Minute Mental Model

This batch is about putting blood or immune cells into a patient safely and for the right reason.

- **Transfusion medicine** asks: What component is missing, what antigen/antibody incompatibility could hurt the patient, and what reaction pattern is happening?
- **Bone marrow transplant** asks: Is the goal marrow rescue from high-dose therapy, immune graft-vs-tumor effect, or cure of a non-malignant hematologic disease?
- **CAR-T and bispecific antibodies** ask: How do we redirect T cells to tumor targets, and how do we manage cytokine release?

The practical exam move is to separate **supportive replacement** (RBCs, platelets, plasma, cryo) from **immune cellular therapy** (alloBMT, CAR-T, bispecifics).

Highest-Yield Concepts

Blood Components and Indications

- RBC transfusion treats severe anemia, often hemoglobin below 7 g/dL. Benefits are restoring volume after rapid blood loss and improving oxygen-carrying capacity. Moderate anemia 7-9 g/dL often lacks benefit and still carries risk. [Transfusion Medicine, slide 5]
- Platelet transfusion thresholds: platelets below 10k for spontaneous bleeding prevention, or below 50k with surgery/invasive procedures. [Transfusion Medicine, slide 6]
- FFP contains coagulation factors II, V, VII, VIII, IX, X, XI, XII, protein C, tissue factor, fibrinogen, vWF, and antithrombin. Use it for coagulopathy with multiple factor deficiencies, such as liver disease, DIC, or warfarin effect; downside is large volume. [Transfusion Medicine, slide 7]
- Cryoprecipitate is derived from plasma and contains fibrinogen, factor VIII, vWF, and factor XIII. It is most often used for low fibrinogen states such as DIC and liver disease. [Transfusion Medicine, slide 8]
- Jehovah's Witness or transfusion refusal cases require clear discussion of benefits, risks of declining, and alternatives such as iron, erythropoietin, treating bleeding source, minimizing phlebotomy, and supportive oxygen/volume strategies. [Workshop, slide 2]

ABO, Rh, and Compatibility Testing

- ABO is the most important RBC antigen system because antigens are dense and antibodies are high-titer. A and B alleles are codominant; O is recessive. [Transfusion Medicine, slide 10]
- Each ABO blood type forms antibodies against absent antigens. RBC transfusions must avoid donor RBC antigens targeted by recipient antibodies. [Transfusion Medicine, slides 11-12]
- Plasma compatibility is reversed logic: plasma transfusion exposes the recipient to donor antibodies. Type AB plasma has no anti-A or anti-B and is broadly compatible as plasma. [Transfusion Medicine, slide 13]
- For a type A recipient, compatible donor RBCs are type A or O; compatible donor plasma is type A or AB. [Transfusion Medicine, slides 14-15]
- Rh/D antigen matters because of immunogenicity, frequent incompatibility, and hemolytic disease of the newborn. [Transfusion Medicine, slide 16]
- Hemolytic disease of the newborn can be prevented with Rh immune globulin given to Rh-negative pregnant/delivering patients to prevent anti-Rh alloimmunization. [Transfusion Medicine, slide 18]
- Compatibility testing proceeds as **type, screen, cross**: determine ABO/Rh, screen for clinically significant unexpected antibodies, then cross-match the specific donor unit. [Transfusion Medicine, slide 19]
- Antibody screen uses indirect Coombs methodology to detect unexpected non-ABO antibodies from alloimmunization or autoantibodies. [Transfusion Medicine, slide 22]

- Crossmatch ensures the specific donor unit is compatible with the recipient. [Transfusion Medicine, slide 23]

Transfusion Reactions

- Acute hemolytic transfusion reaction is immune destruction of donor RBCs by preformed recipient ABO antibodies, most often from clerical/procedural error. It can cause immediate massive hemolysis, shock, renal failure, and DIC. [Transfusion Medicine, slide 24]
- Acute hemolytic reaction symptoms include fever, chills, hypotension, back/flank pain, hematuria, decreased urine output, epistaxis, and oozing from IV lines. [Transfusion Medicine, slide 24]
- Acute hemolysis is mainly intravascular through complement activation; labs show low haptoglobin, plasma free hemoglobin/hemoglobinuria, high LDH, spherocytes, and high bilirubin. [Transfusion Medicine, slide 25]
- Confirm immune-mediated transfusion destruction by repeating donor/recipient typing and DAT on the post-reaction specimen compared with pre-transfusion DAT. Treatment is stop transfusion immediately, IV fluids, and vasopressors as needed. [Transfusion Medicine, slide 26]
- Delayed hemolytic transfusion reaction occurs when prior minor antigen exposure generated an alloantibody whose titer later became undetectable; re-exposure causes an anamnestic response and delayed hemolysis. [Transfusion Medicine, slide 27]
- Transfusion risks include acute/delayed hemolytic reactions, allergic/anaphylactic reactions, febrile nonhemolytic reactions, TRALI, TACO, viral infection, and bacterial sepsis. [Transfusion Medicine, slide 28]

Bone Marrow Transplant Core Concepts

- BMT terms include bone marrow transplantation, blood/marrow transplantation, stem cell transplantation, and hematopoietic cell transplantation. Myeloablative therapy kills normal marrow progenitors; non-myeloablative therapy spares normal marrow progenitors. [BMT, slide 7]
- Allogeneic transplant uses another human donor; autologous uses the patient's own cells; syngeneic uses an identical twin. [BMT, slide 8]
- Donor sources include matched sibling, matched unrelated donor, partially mismatched cord blood, related haploidentical donor, and mismatched unrelated donor. Matched sibling availability is about 25%; haploidentical related donor availability is over 95%. [BMT, slide 8]
- Classical BMT lets clinicians give therapy doses too high for normal marrow tolerance; transplant rescues myeloablation. [BMT, slide 9]
- High-risk AML example: induction treats complications/gets remission; consolidation targets minimal residual disease. Poor-risk disease such as FLT3/ITD+ AML may use allogeneic BMT. [BMT, slides 4-6]
- Allogeneic BMT cure benefit can come from graft-vs-tumor effect, not just chemotherapy dose escalation. [BMT, slides 18-19]
- Autologous BMT avoids GVHD and is most useful when tumors have steep dose-response curves, such as Hodgkin lymphoma and aggressive NHL. [BMT, slide 16]
- Many transplants now use mobilized peripheral blood progenitors collected by apheresis; this yields faster engraftment but also collects more T cells, increasing GVHD risk. [BMT, slide 17]

HLA, GVHD, and Donor Selection

- HLA is the major barrier to transplantation. HLA haplotypes are inherited on chromosome 6; each sibling has about a 25% chance of sharing all major HLA haplotypes. [BMT, slide 14]
- HLA class I antigens A/B/C are recognized by CD8 T cells; class II DR/DP/DQ are recognized by CD4 T cells. DNA PCR typing is now standard. [BMT, slide 14]
- GVHD occurs when the transplanted immune system attacks recipient tissues; common organs are skin, GI tract, and liver. [BMT, slides 8 and 13]
- Even HLA-matched donors can cause GVHD through minor histocompatibility antigens, polymorphic proteins outside the MHC presented by MHC. [BMT, slide 15]
- GVHD is a double-edged sword: it causes morbidity/mortality but correlates with lower relapse because graft-vs-tumor is a major antitumor effect. [BMT, slides 18-20]
- GVHD prevention includes calcineurin inhibitors, sirolimus, T-cell depletion, and post-transplant cyclophosphamide. Active GVHD is treated mainly with corticosteroids and other immunosuppressives. [BMT, slide 21]
- Transplant-related mortality is below 5% for autologous transplant and historically 10-20% for allogeneic, now closer to 5-10%; opportunistic infection from treatment is a common cause. [BMT, slide 22]

- Reduced-intensity/non-myeloablative transplant uses lower-dose immunosuppressive conditioning, can be used in older/less fit patients, and relies primarily on graft-vs-tumor effect. [BMT, slide 23]
- PTCy works because alloreactive T cells are maximally stimulated around days 3-4 after transplant; high-dose cyclophosphamide preferentially targets these cells while memory T cells resist via high ALDH. [BMT, slide 25]
- Increasing mismatch with PTCy does not necessarily worsen GVHD, making haploidentical transplant practical and improving access. [BMT, slide 28]
- Modern donor selection emphasizes timely access and donor age; younger donors often improve engraftment and reduce relapse compared with older donors. [BMT, slide 30]
- In sickle cell disease, mixed chimerism may still be acceptable if donor erythropoiesis reduces HbS enough to prevent disease; in AML, mixed chimerism may suggest relapse risk or inadequate graft-vs-leukemia. [Workshop, slide 5]

CAR-T and Bispecific Antibodies

- Relapsed/refractory hematologic malignancies often have limited chemotherapy options; redirected T-cell therapies create new options: CAR-T cells and CD3-target bispecific antibodies. [CAR-T/BsAbs, slide 3]
- Common targets: CD19 for B-ALL/DLBCL, BCMA for multiple myeloma, and CD20 for B-NHL. [CAR-T/BsAbs, slide 3]
- CAR-T structure incorporates T-cell activation signal 1 through CD3 and signal 2 costimulation such as CD28 into the intracellular signaling domain. [CAR-T/BsAbs, slide 4]
- CAR-T uses the patient's own engineered T cells, requires manufacturing time, and is usually a single infusion with potential months-to-years persistence. [CAR-T/BsAbs, slide 11]
- Bispecific antibodies redirect endogenous T cells, are off-the-shelf, and require continuous or serial dosing. [CAR-T/BsAbs, slide 11]
- CAR-T generally has higher CRS risk and is delivered at certified centers; bispecifics have CRS too, often lower grade, and are expanding into community settings. [CAR-T/BsAbs, slide 11]
- CAR-T relapse can be durable but may occur through antigen escape or loss of CAR-T persistence; bispecific resistance can develop during ongoing therapy. [CAR-T/BsAbs, slide 11]

Cytokine Release Syndrome

- CRS begins when CAR-T or bispecific therapy engages target cells and activates T cells. Activated T cells release IFN-gamma, TNF-alpha, and IL-2; macrophages then release IL-6, IL-1, and IL-10, producing systemic inflammation. [CAR-T/BsAbs, slide 8]
- IL-6 is a key driver of CRS symptoms; tocilizumab blocks the IL-6 receptor. [CAR-T/BsAbs, slide 8]
- CRS grade 1 is fever only: treat with antipyretics and close monitoring. [CAR-T/BsAbs, slide 10]
- CRS grade 2 is fever plus hypotension: treat with tocilizumab. [CAR-T/BsAbs, slide 10]
- CRS grade 3 requires vasopressors or supplemental oxygen: treat with tocilizumab plus steroids. Grade 4 is life-threatening shock or mechanical ventilation: treat with tocilizumab plus steroids. [CAR-T/BsAbs, slide 10]

Must-Know Comparisons

Component	Main contents	Use	Major downside
RBCs	Red cells	Severe anemia, acute blood loss	Hemolytic/allergic/infectious risk
Platelets	Platelets	Platelets <10k or <50k with procedure	Alloimmunization/reactions
FFP	All coagulation factors	Multiple factor deficiency: liver disease, DIC, warfarin	Large volume
Cryoprecipitate	Fibrinogen, VIII, vWF, XIII	Low fibrinogen	Blood product risk

Therapy	Cell source	Timing	Major strength	Major limitation
Autologous BMT	Patient cells	After high-dose therapy	No GVHD	No graft-vs-tumor; tumor contamination possible
Allogeneic BMT	Donor cells	After conditioning	Graft-vs-tumor; curative potential	GVHD, infection, donor logistics
CAR-T	Patient engineered T cells	2-4 week manufacturing	Single infusion, durable potential	CRS, cost, certified centers

Therapy	Cell source	Timing	Major strength	Major limitation
Bispecific antibodies	Patient endogenous T cells	Off-the-shelf	Immediate availability	Ongoing therapy, resistance, CRS

Common Confusions To Avoid

- **RBC compatibility and plasma compatibility run in opposite directions.** RBCs bring donor antigens; plasma brings donor antibodies.
- **Acute hemolytic transfusion reactions are often procedural errors.** Stop the transfusion first.
- **FFP and cryo are not interchangeable.** FFP gives broad factors; cryo is mainly for fibrinogen replacement.
- **Allogeneic BMT is not just marrow rescue.** Much of its antitumor power is graft-vs-tumor.
- **GVHD and graft-vs-tumor travel together.** Less relapse can come at the cost of immune toxicity.
- **HLA matching is important, but modern haploidentical approaches changed access.** PTCy made half-matched donors much more usable.
- **Reduced-intensity transplant still depends on immune effect.** It lowers conditioning toxicity but relies on graft-vs-tumor.
- **CAR-T is not the same as a bispecific antibody.** CAR-T engineers T cells ex vivo; bispecifics redirect existing T cells in vivo.
- **CRS is IL-6-driven clinically.** Tocilizumab is central once CRS reaches grade 2 or higher.

Rapid Review

- RBC transfusion: severe anemia, often Hgb <7; avoid unnecessary transfusion.
- Platelets: <10k prophylaxis; <50k for invasive procedure.
- FFP: multiple coag factor deficiencies; cryo: low fibrinogen.
- Type/screen/cross: ABO/Rh, unexpected antibodies, donor-unit compatibility.
- Acute hemolytic transfusion reaction: ABO mismatch, fever/chills/hypotension/flank pain/hemoglobinuria/DIC; stop transfusion.
- Delayed hemolytic reaction: anamnestic response to minor antigen after prior exposure.
- AlloBMT: donor immune system, GVHD risk, graft-vs-tumor benefit.
- Autologous BMT: no GVHD, dose-escalation logic, useful in Hodgkin/aggressive NHL.
- Reduced-intensity BMT: less conditioning toxicity, relies on graft-vs-tumor.
- PTCy enables haploidentical transplant by deleting alloreactive T cells after transplant.
- CAR-T: autologous engineered cells, single infusion, manufacturing delay.
- Bispecifics: off-the-shelf T-cell redirectors, serial/continuous dosing.
- CRS: fever -> hypotension/oxygen/pressors; tocilizumab blocks IL-6R, steroids for higher grades.

Active Recall Questions

1. When would you transfuse RBCs, platelets, FFP, or cryoprecipitate?
2. Why is unnecessary RBC transfusion harmful?
3. Explain ABO compatibility for RBC transfusion versus plasma transfusion.
4. What is the purpose of type, screen, and crossmatch?
5. How do you recognize and manage an acute hemolytic transfusion reaction?
6. Why can delayed hemolytic transfusion reactions occur after a negative antibody screen?
7. What is the role of Rh immune globulin?
8. Compare allogeneic, autologous, syngeneic, and haploidentical transplant.
9. Why can allogeneic BMT cure malignancy beyond high-dose chemotherapy?
10. What causes GVHD despite HLA matching?
11. Why is GVHD considered a double-edged sword?
12. How does post-transplant cyclophosphamide reduce GVHD?
13. Why might mixed chimerism be acceptable in sickle cell disease but concerning in AML?
14. Compare CAR-T and bispecific antibodies by source, manufacturing, dosing, persistence, and CRS risk.
15. What cytokine pathway does tocilizumab target in CRS?
16. How does CRS grade guide management?